Lab 31: Replacing the auth layer: nobody agrees on the boundary, so plan

Topic 3: Claude Code Configuration & Workflows · Task 3.4: Determine when to use plan mode vs direct execution

Runs on Simulator 2: Code Generation with Claude Code · Run in class

Source: SkillCertPro practice exam Part 1, Q8 (unofficial third-party question)

The question

A payments platform team is replacing its homegrown authorization layer with a policy engine across API handlers, middleware, and tests. Engineers disagree about where the policy boundary should live, and the developer assigned to it is unfamiliar with several affected packages. Which workflow should they choose before implementation?

A. Run a custom slash command that instructs Claude Code to avoid asking questions during the refactor.

B. Use direct execution so Claude Code starts editing immediately and discovers the architecture through test failures.

C. Enter plan mode so Claude Code researches the codebase and proposes changes for approval before editing files.

D. Select the opus model alias and let Claude Code complete the refactor through direct execution.

Simulate the solution

```
# In the browser: open Simulator 2, then Lab 31
cd labs
uv run serve.py

# In Claude Code, from the labs folder
/lab 31

# In a terminal, from simulators/simulator-2-code-generation/
uv run casekit.py --case UC-8
```

What you should see

Files hash the same before and after: a plan, with no edits.

Claude researches src/middleware/authorize.js and its callers and returns a plan; the file-hash table shows zero edits.

Open `simulators/simulator-2-code-generation/evidence/` and find it yourself.

The answer

C. Plan mode researches and proposes before any edit.

Plan mode separates investigation from implementation. It is a documented permission mode in which Claude researches the codebase, may run exploratory shell commands under the normal permission flow, and writes a plan. Source files are not edited until the plan is approved. Anthropic's best practices recommend planning when the approach is uncertain, the change spans multiple files, or the developer is unfamiliar with the code. All three apply here.

Why the others fail

A. Suppressing questions removes exactly the clarification needed when engineers disagree about the design.

B. Discovering an authorization architecture through test failures invites broad, risky rework. Direct execution suits small fixes.

D. A stronger model may reason better, but model choice does not replace the approval boundary that plan mode provides.

Check yourself

- plan mode edited no files
- the output is a plan that names the authorization boundary
- the plan references the real callers found by research

Now change one thing

Set LAB_MODEL=claude-sonnet-5-5, rerun, and compare the result and the cost line in evidence/.

Third-party practice material from SkillCertPro, reconstructed from a video transcript in the Tertiary Markdown Vault. Not a real exam item. Wording may differ from the original.